Patel Asit Sanket

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.73
The H.B. Kapadia New high school, memenagar	Class XII (gujarat secondary and higher secondary education board)	2024	91.23%
The H.B. Kapadia New high school, memnagar	Class X (gujarat secondary and higher secondary education board)	2022	92.67%

SCHOLASTIC ACHIEVEMENTS

- **Specialist** on Codeforces (**max rating: 1553**), regular participant in programming contests (account)
- Solved **600+ questions** across different platforms like Codeforces, Leetcode and Codechef
- Secured **3rd place** in the Lambda Campus Hackathon for innovative problem solution for better campus.

EXPERIENCE

Computer Vision Intern | DRC Systems Pvt. Ltd. *(On-site)*

AI-Powered Golf Swing Analysis -- Python, PyTorch, OpenCV, MobileNetV2, BiLSTM (link)

- Extended a research-grade **SwingNet** implementation into a deployable golf swing sequencing system by architecting modular training, evaluation, and inference pipelines with configurable preprocessing and hardware-aware execution.
- Developed a real-time video analysis pipeline that performs frame-wise swing phase recognition, confidence estimation, automatic event localization, annotated MP4 generation, and event-frame extraction from monocular golf videos.
- Enhanced model generalization through data augmentation and training pipeline improvements, increasing Prediction Correct Events (**PCE**) from **71.5%** (baseline) to **75.6%** while maintaining the original network architecture.
- Integrated the solution into a larger computer vision workflow utilizing **YOLOv8**, face recognition, object tracking, and pose estimation for AI-assisted golf performance analytics.

PROJECTS

xv6 Operating System Extensions -- C (link)

- Worked with MIT's **xv6 OS**, gaining understanding of kernel-user space interaction and syscall flow.
- Designed and implemented a **Multi-Level Feedback Queue (MLFQ)** scheduler with priority boosting and time-slice management to improve scheduling fairness and responsiveness. Developed a **scheduler-aware page replacement policy** by integrating process scheduling behavior with page-replacement decisions.
- Added Disk-simulation in xv6, with support of **RAIDS-0,1,5** protocol for performance and fault tolerance.

RISC-V 5-Stage Pipelined Processor -- C++, Python (link)

- Architected a cycle-accurate **5-stage RISC-V simulator** (IF, ID, EX, MEM, WB) supporting integer-float (RV64F/D) operations. Implemented a robust hazard management unit featuring data forwarding and stalling.
- Implemented **multi-set cache** along with **dynamic branch prediction (1-bit, multi-bit)** for improving performance. Test script in Python for comparing various statistics with different cache and simulator setup.

OpenCV Object Perception & Classification -- Python, OpenCV (link)

- Architected a real-time, CPU-optimized perception pipeline in Python utilizing OpenCV for live video object detection and persistent multi-object tracking using **Hungarian matching** and adaptive **Canny edge detection**. Developed a custom detection pipeline using contour extraction and distance-aware Non-Max Suppression.
- Curated and hand-labeled a custom dataset of 200+ images to train a HOG+SVM classifier. **(86% accuracy)**

Principled OS-Level UNIX Shell -- C (link)

- Engineered a shell from scratch that performs direct process manipulation and process management without high-level library wrappers. Supports arbitrary-length pipelines with correct EOF semantics.
- Implemented **POSIX-compliant I/O redirection**. Implemented interactive line editing via raw terminal control and a **persistent job table** to track and control background/foreground processes.

RELEVANT COURSE

- **Mathematics:** Discrete Maths, Linear Algebra, Applied statistics, Probability and Random Variables, Calculus
- **Computer Science:** Data Structures and Algorithms, Computer Architecture, Software Development Fundamentals, Database and Management Systems, Operating Systems

TECHNICAL SKILLS

- Programming: C/C++ (Experienced with large codebases), Python (Proficient), SQL.

- Competitive Programming: Specialist on Codeforces **(max: 1553),** regular participant in programming contests.

- Tools & DevOps: Git, Docker, Makefile, Jupyter Notebook.

- Libraries: Pytorch, pandas, matplotlib, scikit-learn, OpenCV.

- Concepts: Operating Systems, Computer Architecture, OOP, Data Structures and Algorithms, Machine Learning.

- Data & Quant: Experience with data analysis and feature-engineering and visualization using pandas and matplotlib on real-world datasets; familiarity with derivatives and quantitative finance concepts; worked on alpha creation using **WorldQuant**.

EXTRA CURRICULAR

- Member of the **NSO** Volleyball team; actively participated in campus **volleyball and tennis** tournaments.